ℹ Information about the exam

Read and follow these instructions carefully to increase your chance of getting good marks.

- This is a closed-book exam. You may use a standard English dictionary. Otherwise, **no notes, calculators, mobile phones, or other electronic devices are allowed**.
- Each question awards a number of points, as shown at the end of each question.
- ~~You can "flag" questions (click the small flag at the top of each question) that you e.g. want to review later.~~
- Some questions ask for a justification of your answer; your justification needs to relate the information given in the question, the course content, and your answer.
- Some questions ask you to write code; this code needs to conform to the coding convention. You only need to write function specifications if the question text states so.
- You may use English or Swedish to answer the exam questions.
- Eva and/or Jonas will come to the main exam hall around 9:00 to answer questions (and a bit later to other exam locations).

Consider the following code:

```
const a = 10;

const b = a < 5 ? 13 : 1;

a > b ? 7 : 11;
```

1 Substitution model - result

What is the result of the program on the left?

Answer: 7

Maximum marks: 1

2 Substitution model - evaluation steps

Show all evaluation steps using the **substitution model**, paying particular attention to the order of computation:

```
const a = 10; const b = a < 5 ? 13 : 1; a > b ? 7 : 11;
const b = 10 < 5 ? 13 : 1; 10 > b ? 7 : 11; // replace a = 10 in rest of code:
const b = false ? 13 : 1; 10 > b ? 7 : 11;  // evaluate predicate 10 < 5, which is false
const b = 1; 10 > b ? 7 : 11; // evaluate value of b
10 > 1 ? 7 : 11; // replace b in rest of code
true ? 7 : 11;  // evaluate predicate 10 > 1, which is true
7
```

Maximum marks: 1

Consider the following code:

```
// @precondition m <= n
function mystery_sum(m, n) {
    return m === n
           ? m
           : m + mystery_sum(m + 1, n);
}
```

3 Evaluating recursive functions

What is the result of `mystery_sum(2, 5)` ?

Answer: 14

Maximum marks: 1

4 Iterative processes

Rewrite the function so that it gives rise to an **iterative process**. Your new function should take the same (and the same number of) arguments as `mystery_sum`.

Answer:

```
function mystery_sum(m, n) {
    function sum_iter(s, i) {
        return i === n
               ? s + i
               : sum_iter(s + i, i + 1);
    }
    return sum_iter(0, m);
}

or

function mystery_sum(m, n) {
    function sum_iter(s, m, n) {
        return m > n
               ? s
               : sum_iter(s + m, m + 1, n);
    }
    return sum_iter(0, m, n);
}

(or similar variations)
```

Maximum marks: 2

5 Variants

Give a **variant** for the (original) function `mystery_sum` as given on the left:

Answer:

n - m

Justify your answer, i.e. why is it a valid variant?

The expression is one over the function's inputs. By the precondition and the base case, the expression will never become negative, and it decreases by 1 in each recursive call.
Thus the expression satisfies the two conditions for being a valid variant.

Maximum marks: 2

6 Higher-order functions

Consider the following function definitions:

```
function plus_one(x) {
    return x + 1;
}
function trans(f) {
    return x => 2 * f(2 * x);
}
function twice(f) {
    return x => f(f(x));
}
```

What is the result of `trans(plus_one)(1)` ?

Answer: 6

What is the result of `trans(twice(plus_one))(1)` ?

Answer: 8

Maximum marks: 2

7 List principle

Consider the following two lists:

```
const a = list(1, 2, 3);
const x = list("e", list("g", "h"), "f");
```

What is the result of `head(tail(tail(x)))` ?

Answer: "f"

What is the result of `length(pair(a, x))` ?

Answer: 4

Maximum marks: 2

8 List processing

Write a **full** function specification for the following function, including **at least two examples** that explain the behavior of the function.

```
function lal(lst, k, val) {
    // variant: k
    return k === 0
                ? null
            : is_null(lst)
                ? pair(val, lal(lst, k - 1, val))
            : pair(head(lst), lal(tail(lst), k - 1, val));
}
```

Your specification should have a **precondition**.

Answer:

```
Example solution (renaming of the function is not needed):

/**
 * Given a list, a desired length and a value, create a list that is like
 * lst but with length exactly k. If k is less than the length of lst,
 * the result has the first k values of lst. If k is greater than the
 * length of lst, copies of the value are added to get length k.
 * @example
 * list_adjust_length(list(), 3, 0); // returns list(0, 0, 0)
 * list_adjust_length(list(1, 2), 4, -4); // returns list(1, 2, -1, -1)
 * @param {list} lst - A list
 * @param {number} k - A desired length
 * @param val - A value for padding (or any type)
 * @precondition k >= 0
 * @returns {list} A list of length k with elements from lst, possibly
 *                    padded with copies of val.
 */
function list_adjust_length(lst, k, val) {
    return k === 0
            ? null
            : is_null(lst)
            ? pair(val, list_adjust_length(lst, k - 1, val))
            : pair(head(lst), list_adjust_length(tail(lst), k - 1, val));
}
```

Maximum marks: 3

9 List processing abstractions

Write the function `sum_even(lst)` that takes as input a list of numbers `lst` and returns the sum of all even numbers in `lst`.

For example:

```
sum_even(list(1, 3, 3, 2, 5, 8, 9, 11, 4)); // returns 14
sum_even(list());         // returns 0
sum_even(list(1, 3));     // returns 0
sum_even(list(1, 2, 3)); // returns 2
```

You **must** use the higher-order list processing functions `map` , `filter` and/or `accumulate` to implement your function (choose one or more that are appropriate). You are **not** allowed to write a recursive function or to use `for` or `while` loops.

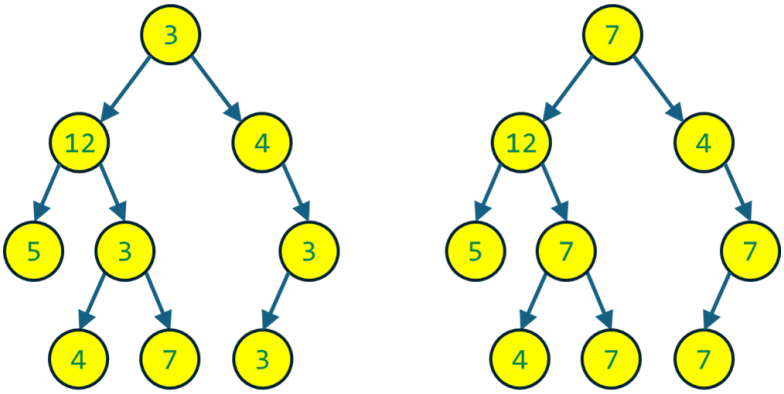
Answer:

```
function sum_even(lst) {
    const even = filter(x => x % 2 === 0, lst);
    return accumulate((x, acc) => acc + x, 0, even);
}
```

Maximum marks: 3

10 Binary Tree Replace Value

Define a function `binary_tree_replace_value` that takes as parameters a binary tree `t`, a value `u`, and another value `v`. The result of `binary_tree_replace_value(t, u, v)` should be a new binary tree that is like `t` except that any node value `u` has been replaced by `v`. For example, suppose that `t` is the tree on the left in the figure, then the result of `binary_tree_replace_value(t, 3, 7)` would be the tree on the right in the figure.



Assume your program already contains

```
import { entry, is_empty_tree, is_tree, left_branch,
         make_empty_tree, make_tree, right_branch} from "binary_tree";
```

Here is a summary of the binary tree interface if you need it: Binary Tree API.

Answer:

```
function binary_tree_replace_value(t, u, v) {
    return is_empty_tree(t)
                ? make_empty_tree()
            : make_tree(entry(t) === u
                        ? v : entry(t),
                        binary_tree_replace_value(left_branch(t), u, v),
                        binary_tree_replace_value(right_branch(t), u, v));
}
```

Maximum marks: 3

11 Merge sort

Recall the **merge sort** sorting algorithm. Suppose the algorithm is applied to the following list:

```
list(5, 11, 2, 7, 3, 5, 15, 1)
```

What are the arguments to the immediate recursive calls of the merge sort algorithm (i.e. to the top-level `merge_sort` function)?

Answer:

The arguments are:

list(5, 11, 2, 7), list(3, 5, 15, 1)

Explain your answer:

The merge sort algorithm proceeds by splitting the input list repeatedly in half, and continuing to sort the smaller lists. Once the smaller lists are sorted, they will be merged into an overall sorted list.

Maximum marks: 2

12 Set data abstraction

At a high level, what is the difference between a **list** and a **set** data abstraction?

Do not discuss complexity or implementation details.

Answer:

In a list, items are stored in a given order, and may contain (explicitly) duplicate items. For sets each item is either present or not, i.e. there are no duplicates and there is no inherent ordering between the items.

Maximum marks: 1

Consider the following program:

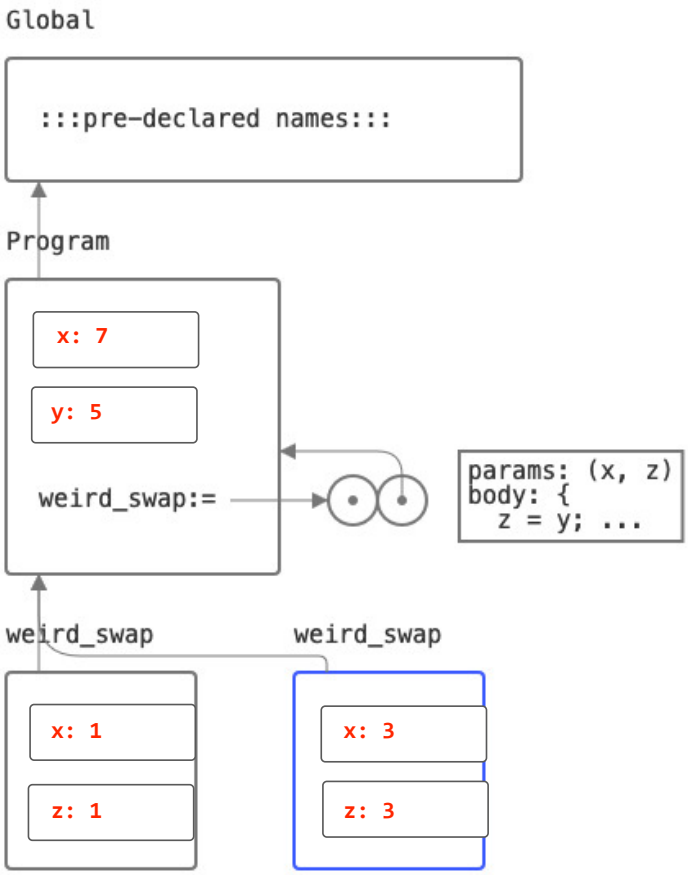
```
let x = 7;
let y = 1;

function weird_swap(x, z) {
    z = y;
    y = x;
    x = z;
}

weird_swap(3, 0);
weird_swap(5, 1);
```

13 Environment model

Fill in the missing entries in the following environment model illustration as they appear **at the end of the program**.
There are **6 blanks** in total that you should fill in. For each blank, use the format `name: value`.



Maximum marks: 3

14 Array Left Scan

Write a function `array_left_scan(f, i, a)` that takes as input a function `f`, an initial value `i`, and an array `a` of length `k` and that returns a new array `b` of length `k`, where `b[0]` is `f(i, a[0])`, `b[1]` is `f(f(i, a[0]), a[1])`, …, and `b[k-1]` is `f(f(… f(f(i, a[0]), a[1]) …, a[k-2]), a[k-1])`.
For example, the result of

```
array_left_scan((x, y) => x * y, 1, [2, 4, 6, 8, 10]
```

should be

```
[2, 8, 48, 384, 3840
```

as $1 \times 2 = 2$, $2 \times 4 = 8$, $8 \times 6 = 48$, $48 \times 8 = 384$, and $384 \times 10 = 3840$.

You may use recursion or iteration (i.e., a for or a while loop). The function should **not** change the input array `a`.
Answer:

```
function array_left_scan(f, i, a) {
    const b = [i * a[0]];
    for (let j = 1; j < array_length(a); j = j + 1) {
         b[j] = f(b[j - 1], a[j]);
    }
    return b;
}

alternatively:

function array_left_scan(f, i, a) {
    const b = [];
    for (let j = 0; j < array_length(a); j = j + 1) {
         b[j] = f(i, a[j]);
         i = b[j];
    }
    return b;
}
```

Maximum marks: 3

15 Counting Change

Recall the problem of counting change from multiple lectures and SICP JS. As stated, given an amount of currency (e.g., in Swedish krona) and a set of denominations (e.g., 1 krona, 2 krona, 5 krona and 10 krona), the function `count_change` computes the number of different ways that the amount could be delivered using these denominations. For example, 5 krona could be delivered in four ways: five 1 krona, two 2 krona and two 1 krona, one 2 krona and three 1 krona, or five 1 krona.

This is the code:

```
function count_change(amount) {
    return cc(amount, 4);
}
function cc(amount, kinds_of_coins) {
    // variant: amount + kinds_of_coins + 10
    return amount === 0
           ? 1
           : amount < 0 || kinds_of_coins === 0
           ? 0
           : cc(amount, kinds_of_coins - 1) +
               cc(amount - first_denomination(kinds_of_coins), kinds_of_coins);
}
function first_denomination(kinds_of_coins) {
    return kinds_of_coins === 1 ? 1
         : kinds_of_coins === 2 ? 2
         : kinds_of_coins === 3 ? 5
         : kinds_of_coins === 4 ? 10
         : 0;
}
```

A call might look like `count_change(100)`.

Could `cc` (and thus `count_change`) benefit from **memoization**? Why? Or why not?

Answer:

Format

The cc function will be called many times with the same arguments. It thus will benefit from memoization to store and avoid recomputing the same recursive calls more than once (by retrieving the results from an array).

Words: 0

Maximum marks: 2

16 Order of Growth

Let $f(n) = n^2 + 3n - 4$

Which of the following statements are true? Select all, i.e. add a check mark for all that are true.

- [x] $f(n) \in O(n^3)$
- [x] $f(n) \in \Omega(n \log n)$
- [x] $f(n) \in \Theta(n^2)$
- [ ] $f(n) \in \Omega(n^3)$

Maximum marks: 2

Muster theorem

Let $T(n)$ be a function such that for some $a, b, c, n_0 > 0$,

$$T(n) = \begin{cases} c, & \text{if } n \leq n_0; \\ aT(n-b) + f(n), & \text{if } n > n_0. \end{cases}$$

If $f(n) \in \Theta(n^d)$ with $d \geq 0$, then

$$T(n) \in \begin{cases} O(n^d a^{n/b}), & \text{if } a > 1; \\ \Theta(n^{d+1}), & \text{if } a = 1; \\ \Theta(n^d), & \text{if } a < 1. \end{cases}$$

Master Theorem

Let $T(n)$be a function such that for some $a, b, c, n_0 > 0$,

$$T(n) = \begin{cases} c, & \text{if } n \leq n_0 \\ aT(n/b) + f(n), & \text{if } n > n_0 \end{cases}$$

Let $\rho = \log_b a$. Then

$$T(n) \in \begin{cases} \Theta(n^\rho), & \text{if for some } \epsilon > 0, \text{ we have } f(n) \in O(n^{\rho-\epsilon}); \\ \Theta(n^\rho \log^{k+1} n), & \text{if for some } k \geq 0, \text{ we have } f(n) \in \Theta(n^\rho \log^k n); \\ \Theta(f(n)), & \text{if for some } \epsilon > 0, \text{ we have } f(n) \in \Omega(n^{\rho+\epsilon}) \text{ and } f \text{ is regular.} \end{cases}$$

17 Solving Recurrences

Write the following recurrence in *closed form*, i.e., without recursion.

$$T(n) = \begin{cases} 1, & \text{if } n \leq 1; \\ 7T(n-3) + n, & \text{if } n > 1. \end{cases}$$

You should use the Muster or the Master theorem to solve the recurrence (see definition on the left).

State the solution of the recurrence and **explain** how you obtained it: explain which theorem you used and why, state what values the theorem's parameters have, which case applies and how you obtain the final solution.

Answer:

Format

We use the Muster theorem, because the recurrence is a decremental one.

a = 7, b = 3, c = 1
f(n) = n which is in Theta(n) with d = 1

Since a = 7 and 7 > 1, the first case of the theorem applies.

Hence T(n) \in O(n^d a^(n/b)) = O(n * 7^(n/3))

Words: 0

Maximum marks: 2

18 Design methodology

Suppose you want to write a program that manages Internet bookmarks. For each bookmark, you want to store a name (e.g., Uppsala University), a URL (e.g., https://www.uu.se/), and the number of times the bookmark has been accessed (e.g., 42).

Define the data type *Bookmark* as we have done in the course. Your data type must have at least one **invariant**.

Give 4 meaningful **examples**, including valid, invalid and borderline/corner cases.

Answer:

```
/*
 * A {Bookmark} is an array[string, string, number] with three items:
 * the first string stores the name of a webpage, the second string
 * stores the webpage's URL, and the final number stores the is_number
 * of accesses.
 * Invariant: the second string is not empty, and the number of access is
 *            non-negative
 */

const bookmark_valid = ["Uppsala University", "https://www.uu.se/", 42];
const bookmark_invalid = ["Uppsala University", "https://www.uu.se/", -42];
const bookmark_corner = ["Uppsala University", "https://www.uu.se/", 0];
const bookmark_corner2 = ["", "foo.bar", 0];
```

Maximum marks: 3

19 Streams

Consider the following program:

```
function stream_map(fun, stream) {  // same as in the lecture
    return is_null(stream)
           ? null
           : pair(fun(head(stream)),
                  () => stream_map(fun, stream_tail(stream)));
}
function numbers(i) {
    display(i);
    return i > 10
           ? null
           : pair(i, () => numbers(i + 1));
}
const doubles = stream_map(x => x * 2, numbers(1));
display("---");
head(stream_tail(doubles));
display("---");
head(stream_tail(stream_tail(doubles)));
```

Which values get printed by the **display** statements (in italic) when the program is evaluated? Give the **concrete values** in the order that they appear and also **explain your answer.**

Answer and justification:

1
"---"
2
"---"
2
3

(6) // OK if added, but is not printed by display

Maximum marks: 2

The definition of const doubles will evaluate the first element of the stream, that is, it will compute 2 * 1, where the 1 is the first element of the numbers stream which will print 1. It will not evaluate more elements, so only 1 is printed before the first "---".

head(stream_tail(doubles)); will evaluate the second element of the stream_map, which in turn evaluates the second element of numbers, hence printing 2 and the "---".

head(stream_tail(stream_tail(doubles))); will force the computation of the first three elements of numbers, and will print 2 and 3. It won't print 1, because this is computed (once) when the numbers(1) stream is defined.